

Nikita Vostrosablin, PhD

Engineering Leader focused on production ML and backend systems. I bring experience from both research-heavy and product-focused environments and remain deeply technical while driving operational improvements.

PRAGUE, CZECH REPUBLIC

CONTACT

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EXPERIENCE

Stats Perform

Engineering Manager | Jul 2025 - Present | Prague, Czechia

Lead a team of 9-12 engineers (predominantly mid-level, ~20% senior) owning production ML and data pipelines that support sports analytics products used by some of the world's biggest broadcasters and professional clubs.

SCOPE & SYSTEMS

- Own architecture, reliability, and stability of ~30 production ML and data pipelines and ~150 distributed Python services running on AWS and on-prem Kubernetes clusters.
- Drive architectural decisions for new systems and product launches, while improving the reliability and operability of existing production systems.
- Own end-to-end production health, incident response, and on-call operations across all owned services.

SCALE & PERFORMANCE

- Systems operate across both high-throughput pipelines processing thousands of events/sec and streaming workloads running at 25 Hz (video broadcast frame rate).
- *Example of a key achievement:* Reduced processing time in key pipelines by up to ~80% via architectural redesign and algorithmic optimization (e.g., eliminated Kafka single-topic bottlenecks by introducing fan-out architectures across services)

RELIABILITY & OPERABILITY (CONTRIBUTION EXAMPLES)

- Defined and enforced pipeline-level SLOs (latency and data completeness), e.g.:
 - metric generation every minute with ≤ 5 min latency
 - $\geq 95\%$ successful batch completion rate
- Introduced structured observability (metrics, alerting, on-call processes), reducing:
 - incident resolution time from multiple days $\rightarrow$ ~1 day
 - support allocation from ~25% $\rightarrow$ ~2% of team capacity
- Led cross-team incident resolution for systemic failures, identifying upstream bottlenecks and driving organization-level support process changes

ARCHITECTURE & DESIGN

- Continuously contributed to the architectural design of multiple systems, both within the team and across team boundaries, shaping pipelines, and services designs across AWS/cloud and on-prem data center environments.
- *Example of contribution:* Influenced cross-team technology choices during the design of a shared streaming system by identifying Apache Flink as a better fit than the sub-optimal technologies used in the initial PoC, educating 3 collaborating teams on the trade-offs, and helping drive adoption; the pipeline was later rebuilt with Flink and confirmed by engineers as the stronger solution.

AI/ML PLATFORM INTEGRATION

- Supported both online and offline inference workflows, ensuring pipeline integration and latency constraints (~minutes-scale SLA per game segment)
- Partnered with AI teams owning model evaluation; responsible for reliability and scalability of inference pipelines

EXECUTION & DELIVERY

- Established CI/CD and service standardization enabling multiple deployments per day
- Reduced idea-to-production lead time to ~1 week on average
- Improved delivery velocity through platform standardization, documentation, and automation initiatives

LEADERSHIP & ORG IMPACT

- Hired 3 engineers directly; contributed to dozens of hiring processes across multiple teams
- Led 3 promotions and managed underperformance through structured feedback and coaching
- Reorganized team into squad-based ownership model, improving focus and accountability, and introduced the DX4 framework to track team performance through a combination of automated metrics collection and team surveys.
- Initiated and led RQDX program (Reliability, Quality, Developer Experience):
 - reduced technical debt
 - standardized services
 - improved developer productivity and system reliability

CROSS-TEAM IMPACT

- Systems serve as critical dependencies for multiple downstream teams and products
- Core contributor to multi-team architecture design for a large-scale streaming platform

AI ADOPTION

- Led AI adoption in the team by treating AI as a tool rather than a shortcut: focused first on preparing the engineering environment, workflows, and guardrails needed to avoid "garbage in, garbage out," while keeping human review and control over generated outputs.
- Drove adoption through education, practical enablement, and result-based work with skeptics, moving the team from near-zero usage to a state where ~75-80% of code is generated by agents under controlled workflows.

Senior Software Engineer, Team Lead | Dec 2023 - Jul 2025

Acted as technical lead for a 4-engineer sub-team responsible for productionizing ML pipelines and distributed systems for the Opta Vision platform, delivering AI-driven metrics across multiple competitions.

KEY CONTRIBUTIONS

- Designed and implemented multiple event-driven pipelines and services generating real-time and post-game metrics
 - (Scale of the systems is described in the EM section above)
- Rebuilt underperforming pipelines, removing architectural bottlenecks and improving throughput
- Identified and fixed critical inefficiencies in DynamoDB usage:
 - redesigned schema to eliminate expensive scan operations
 - achieved significant cost reduction and performance improvement
- Led migration of compute workloads from AWS-managed services to Kubernetes as part of a cost optimization initiative

TECHNICAL LEADERSHIP

- Drove architecture decisions
- Mentored engineers
- Introduced design-first development (design docs), improving alignment and reducing rework
- Participated in hiring processes

Merck (MSD Czech Republic)

Scientific Software Engineer, Team Lead | Jul 2021 - Nov 2023 | Prague, Czechia

- Developed advanced algorithms for scientific and bioinformatics applications
- Contributed to a patented reaction-pathfinder system (US Patent: 20240105285)
- Built AWS-based, event-driven genomics platforms with Batch, Lambda, Step Functions, SQS, DynamoDB, and Redshift
- Implemented Terraform-based infrastructure, CI/CD, and production observability practices
- Led cross-functional delivery of a production mRNA optimization platform

Python Developer / Machine Learning Engineer | May 2019 - Jul 2021

Worked in the AI group across business domains and collaborated with bioinformatics teams on molecular computation and optimization challenges for drug discovery.

Dimension Data

Data ETL Engineer | Dec 2017 - Apr 2019 | Czech Republic

- Built BI infrastructure from scratch for a major European client
 - Designed automated dataflows and operational reporting workflows
 - Maintained data quality and stakeholder-facing delivery
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ORIFLAME SOFTWARE

Intern - Data Scientist, BI/DWH Developer | Jun 2017 - Aug 2017 | Olomouc

- Applied ML and NLP for sales forecasting and customer-review sentiment analysis
 - Built ETL connector pipelines with Pentaho ETL and Python automation
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Freelance

Freelance Web Developer | Sep 2016 - Dec 2016

Built web tooling and automated mail reporting for a renovation-oriented web service using Node.js, HTML, CSS preprocessors, and template engines.

DTU - Technical University of Denmark

Researcher | Sep 2016 - Nov 2016 | Kongens Lyngby, Denmark

Worked on quantum optomechanics research with focus on theoretical foundations of a stroboscopic optomechanical transducer in collaboration with an experimental research team.

EDUCATION

- **Univerzita Palackeho v Olomouci**
PhD, Quantum Physics (Sep 2014 - Mar 2021)
- **Lomonosov Moscow State University (MSU)**
Master of Science, Physics (2008 - 2014)

LANGUAGES

- Russian (Native)
 - English (Fluent, Professional Working)
 - French (Limited Working)
 - Czech (Intermediate)
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TECHNOLOGIES

- **Languages & Core:** Python, Go, SQL, Bash, C++, JavaScript
- **Cloud & Infrastructure:** AWS, Kubernetes, IaaS (Terraform), CI/CD (Jenkins, ArgoCD)
- **Streaming & Messaging:** Kafka, Apache Flink, SQS, SNS
- **Databases:** Relational (Redshift, PostgreSQL, MySQL), NoSQL (DynamoDB), Graph (Neo4j)
- **ML / Data Science:** pandas, Polars, scikit-learn, PyTorch
- **Orchestration:** Airflow, Temporal
- **Observability:** ELK, Grafana, OpenTelemetry

SELECTED PUBLICATIONS

- Statistical spring in the Advanced LIGO detector with unbalanced arms and in the Michelson-Sagnac interferometer
- mRNAid, an open-source platform for therapeutic mRNA design and optimization strategies
- Pulsed quantum interaction between two distant mechanical oscillators
- Squeezer-based pulsed optomechanical interface
- Pulsed quantum continuous-variable optoelectromechanical transducer